

WASP-36b

R-1881

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_191115 · created 2026-08-03T11:49:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458802.9171 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.048
Transit depth (Rp/Rs)^2	2.78 +/- 1.59 [%]
Orbital Inclination (inc)	86.06 +/- 2.94
Airmass coefficient 1 (a1)	0.76 +/- 0.46
Airmass coefficient 2 (a2)	-0.01 +/- 0.12
Scatter in the residuals of the lightcurve fit is	60.17 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	6.75
Transit Duration (day)	0.088 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.048 vs 0.1368 (deviation 0.62σ)
Duration fit vs published	0.088 d vs 0.0758 d (deviation 0.86σ)
Mid-time O–C	-1.0 min
Depth SNR	3.4
Residual scatter	60.17 %

Light curve

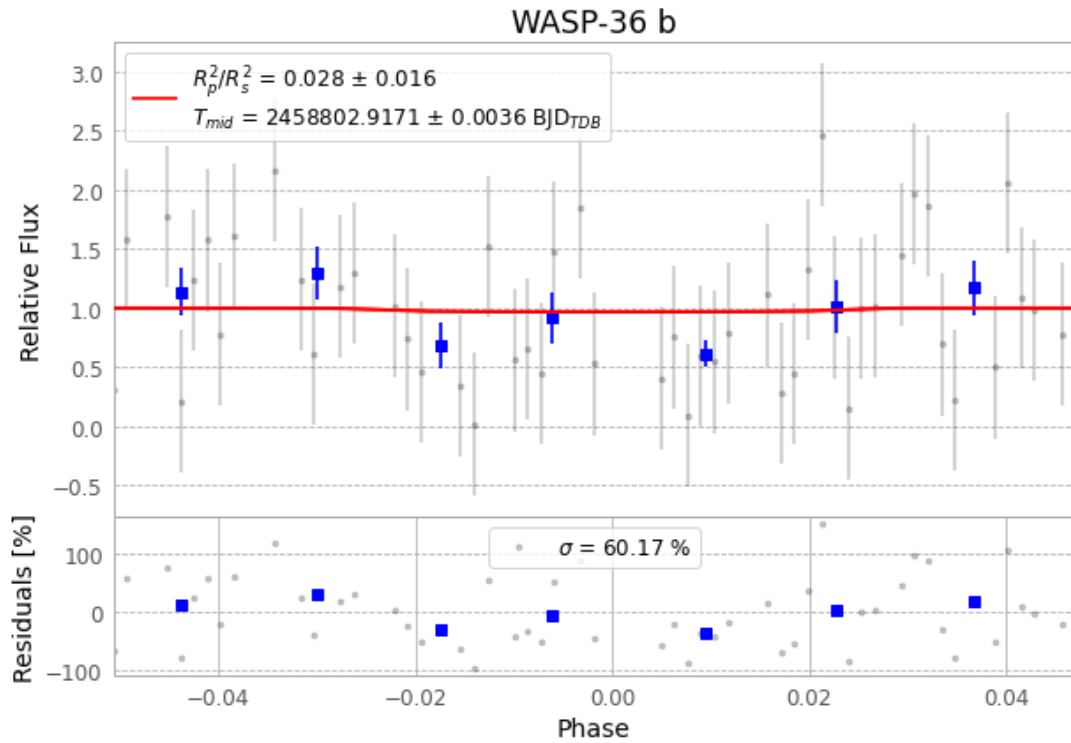


Figure 1 — FinalLightCurve_WASP-36 b_2019-11-15.png (SHA-256 1298a104e7e613f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [214, 379], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7c45abc9e8a79643...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-36191115080612.FITS → WASP-36191115114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-191115.log (f3ce476023e9...), FinalLightCurve_WASP-36 b_2019-11-15.png (1298a104e7e6...), FinalLightCurve_WASP-36 b_2019-11-15.csv (8ea9cd6afe2c...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:49Z.